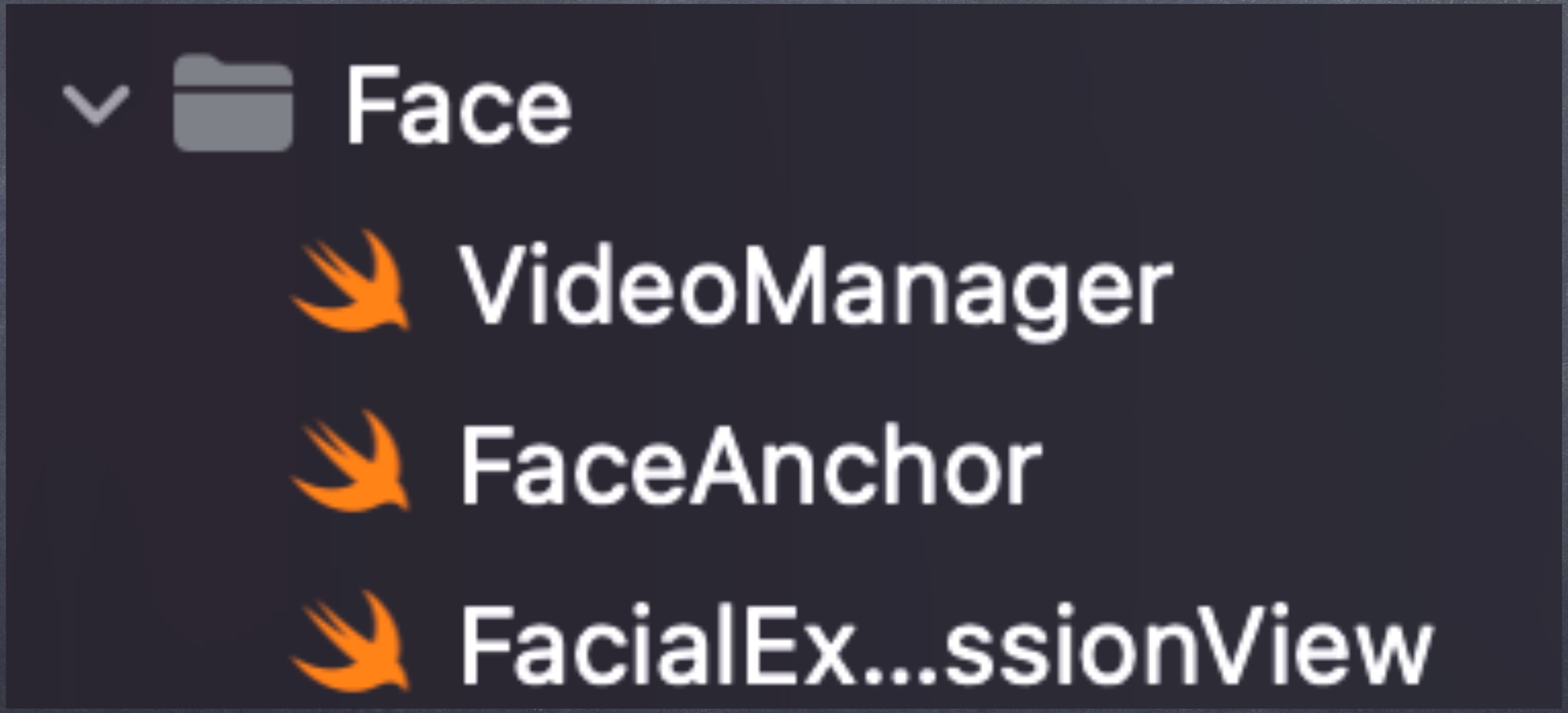
스누즈 – 사용자 감정케어 앱

박유진, 홍시은, 김도현

iOS AR Face Tracking



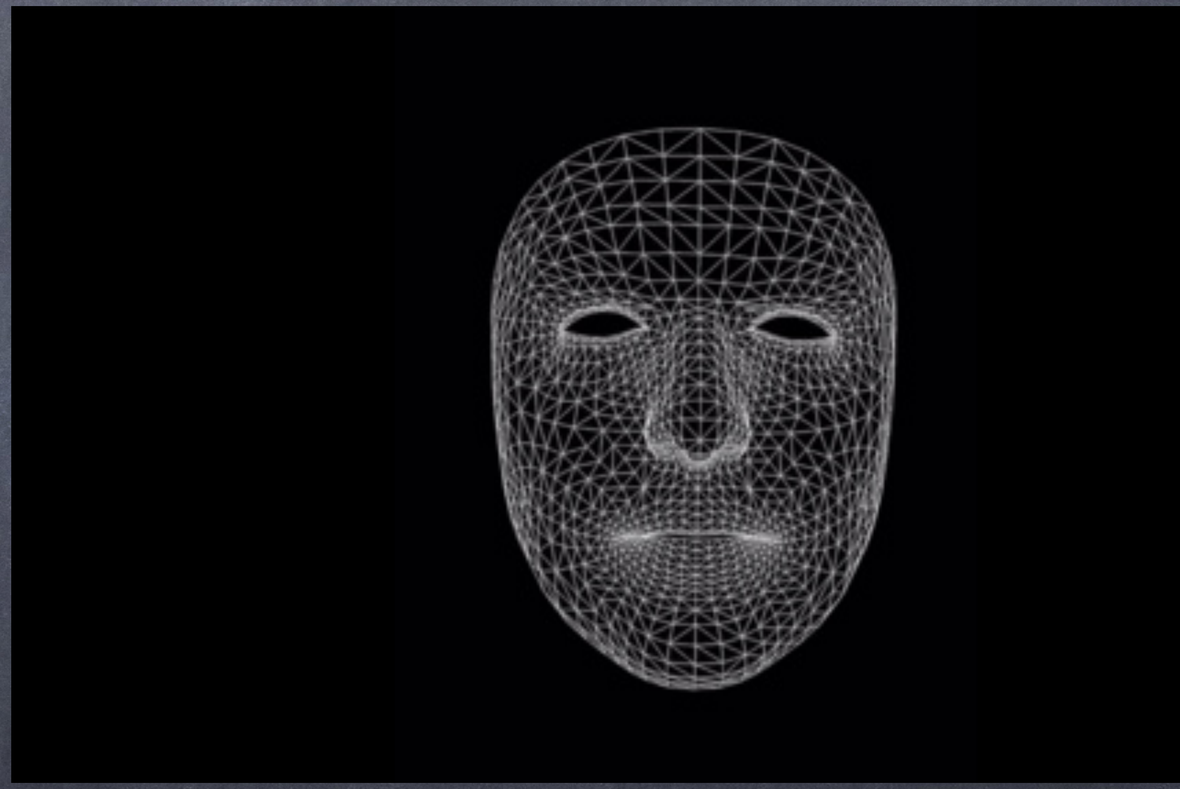
VideoManager 코드의 일부

```
import AVKit
import SwiftUI

protocol VideoManagerProtocol: AnyObject {
    func didReceive(sampleBuffer: CMSampleBuffer)
}

final class VideoManager: NSObject {
    @EnvironmentObject var emotionManager: FaceEmotionManager

    weak var delegate: VideoManagerProtocol?

    private let captureSession = AVCaptureSession()
    private let videoDevice = AVCaptureDevice.default(for: .video)

    lazy var videoLayer: AVCaptureVideoPreviewLayer = {
        return AVCaptureVideoPreviewLayer(session: captureSession)
    }()

    private lazy var videoOutput: AVCaptureVideoDataOutput = {
        let output = AVCaptureVideoDataOutput()
        let queue = DispatchQueue(label: "VideoOutput", attributes: DispatchQueue.Attributes.concurrent, autoreleaseFrequency:
            DispatchQueue.AutoreleaseFrequency.inherit)
        output.setSampleBufferDelegate(self, queue: queue)
        output.alwaysDiscardsLateVideoFrames = true
        return output
    }()
```

카메라 영상처리와 관련하여서 카메라의 영상 프레임을 받아오는 파일

FaceAnchor 코드의 일부

```
import ARKit

protocol FaceAnchorDelegate: AnyObject {
    func update(expression: String)
}

final class FaceAnchor: NSObject {
    weak var delegate: FaceAnchorDelegate?

    private var expression = ""

    func analyze(faceAnchor: ARFaceAnchor) {
        mouth(faceAnchor)
        eyebrow(faceAnchor)
        tongue(faceAnchor)
        mouthAndeyes(faceAnchor)
    }

    private func mouth(_ faceAnchor: ARFaceAnchor) {
        let mouthSmileLeft = faceAnchor.blendShapes[.mouthSmileLeft] as? CGFloat ?? 0
        let mouthSmileRight = faceAnchor.blendShapes[.mouthSmileRight] as? CGFloat ?? 0
        let smile = (mouthSmileLeft + mouthSmileRight) / 2
        DispatchQueue.main.async { [weak self] in
            self?.isSmile(value: smile)
        }
    }
```



출처: https://developer.apple.com/videos/play/tech-talks/601/

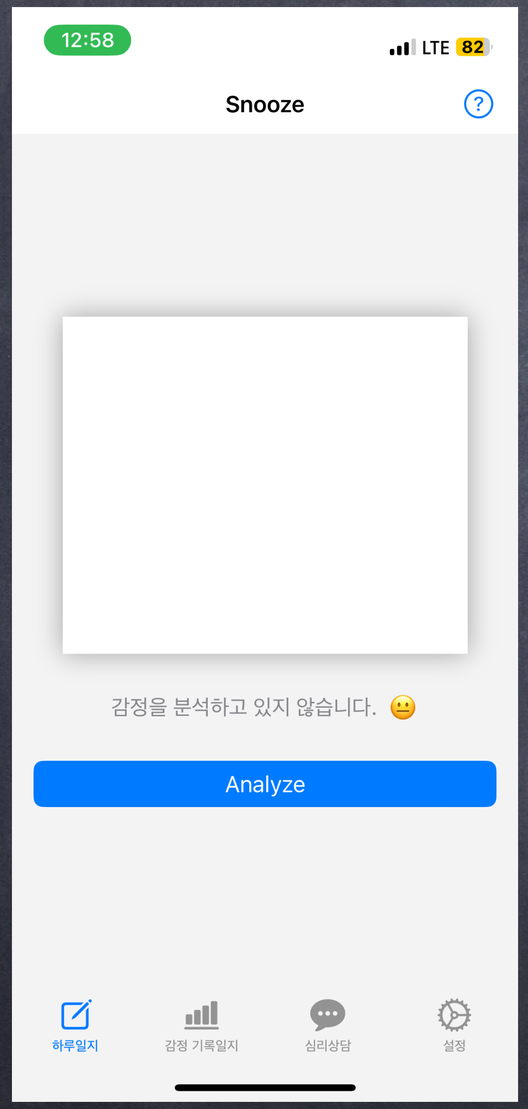
받아온 카메라 영상 프레임에서 얼굴의 특징점들을 추출해서 표정을 판별하는 파일

FacialExpressionView 코드의 일부

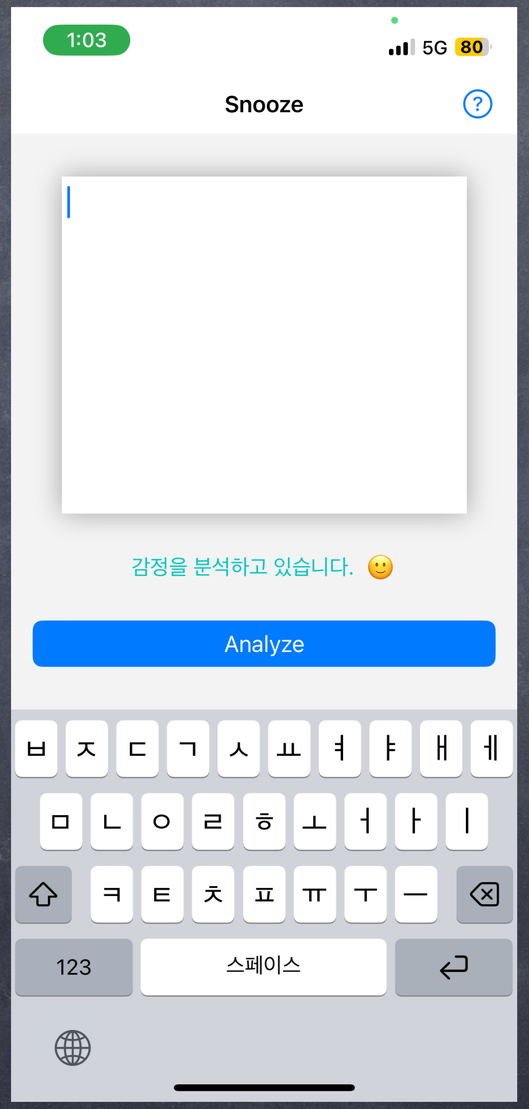
```
extension FacialExpressionController: ARSessionDelegate, FaceAnchorDelegate {
    func session(_ session: ARSession, didUpdate anchors: [ARAnchor]) {
        for anchor in anchors {
            if let faceAnchor = anchor as? ARFaceAnchor {
                face.analyze(faceAnchor: faceAnchor)
            }
        }
    }

    func update(expression: String) {
        self.expression = expression
        self.expressionsOfRecognized.insert(expression)
        setEmotion(expression)
    }
}

struct FacialExpressionViewRefer: UIViewControllerRepresentable {
    @EnvironmentObject var emotionManager: FaceEmotionManager

    @Binding var expression: String
    @Binding var expressionsOfRecognized: Set<String>

    func makeUIViewController(context: Context) -> FacialExpressionController {
        return FacialExpressionController(expression: $expression, expressionsOf
            emotionManager.setEmotion(emotion)
        }
    }
```

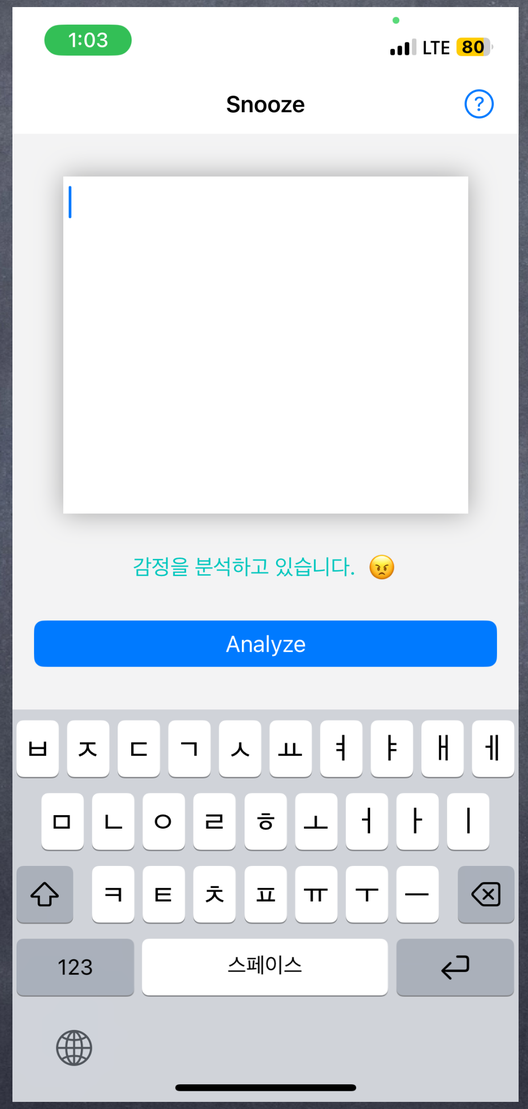
앞에서 정의한 코드들을 사용하여서 카메라 영상 프레임을 ARFaceAnchor에 넘겨주고 얼굴표정을 판별하도록 View를 정의한 파일

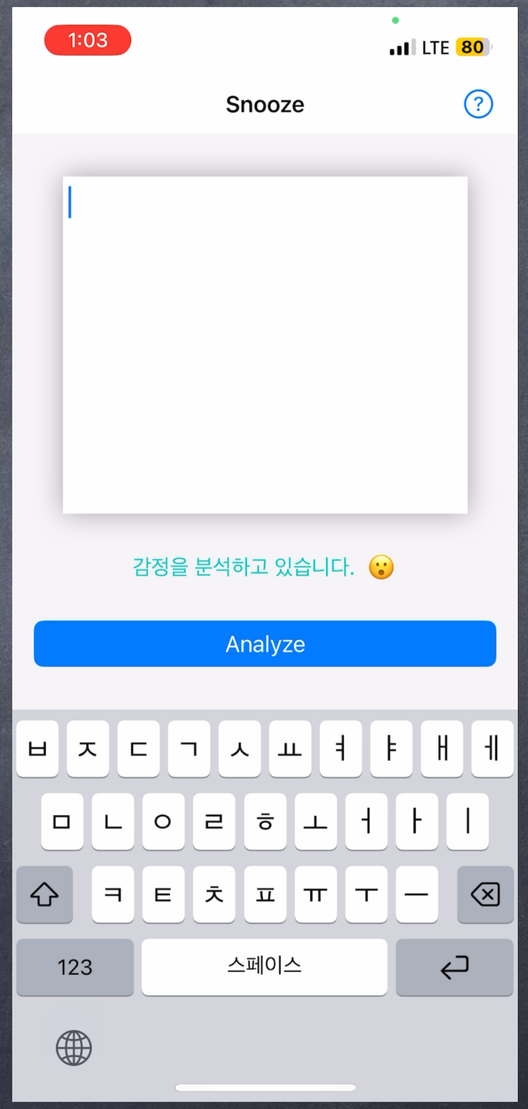
실행화면


12:58
LTE 82
Snooze
감정을 분석하고 있지 않습니다. 😐
Analyze
하루일지
감정 기록일지
심리상담
설정

입력전


1:03
5G 80
Snooze
감정을 분석하고 있습니다. 🙂
Analyze
123
스페이스


1:03
LTE 80
Snooze
감정을 분석하고 있습니다. 😠
Analyze
123
스페이스


1:03
LTE 80
Snooze
감정을 분석하고 있습니다. 😮
Analyze
123
스페이스

입력중